Coulomb's law

→ Charge is an intrinsic property of a particle.

↳ Similar to how mass is an intrinsic property that determines how an object responds to gravitational interactions, Electric charge determines how a particle responds to electromagnetic interactions.

→ Two charges: positive / +ve & negative / -ve

→ Charge is quantized $q = \pm Ne$ where e is charge of an electron & $e = 1.60218 \times 10^{-19} C$.

→ Electric charge is conserved in an isolated system.

→ Like charges repel & unlike charges attract.

→ Electron transfer between materials gives them charge.

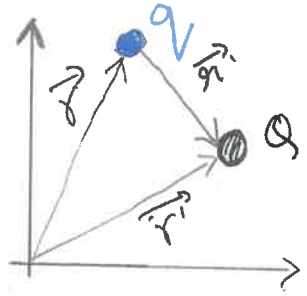
Coulomb's law:

→ Force on a test charge Q due to a single Point charge q that is at a distance r away is given by

$$\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{qQ}{r^2} \hat{r}$$

ϵ_0 is called true Permitivity of free space

$\epsilon_0 = 8.85 \times 10^{-12} \frac{C^2}{Nm^2}$



$$\vec{R} = \vec{r}' - \vec{r}$$

$$\hat{R} = \frac{\vec{r}' - \vec{r}}{|\vec{r}' - \vec{r}|}$$

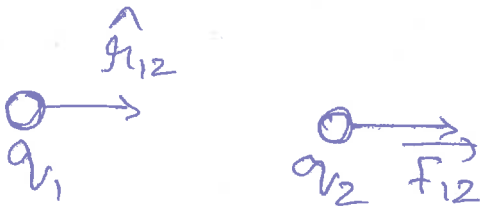
} Recall this



$$\vec{F}_{12} = \frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{R^2} \hat{R}_{12}$$

Vector representation of Coulomb's law.

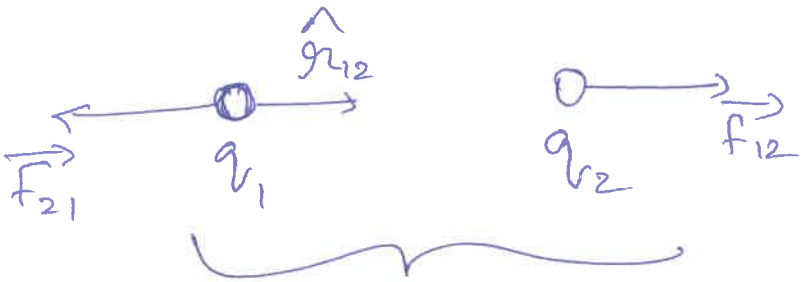
If q_1 & q_2 are both +ve



q_2 will experience $\vec{F}_{12}$ in direction of $\hat{R}_{12}$.

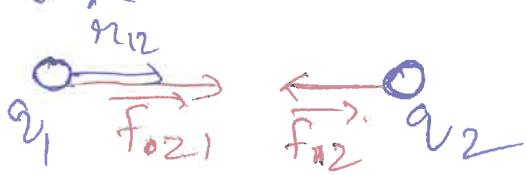
Now applying Newton's 3rd law $\vec{F}_{12} = -\vec{F}_{21}$

We get the force direction on F_1



This shows that like charges repel.

If they are opposite charges



} they attract.

→ Now if there are several point charges: $q_1, q_2, q_3 \ldots q_n$ at $r_1, r_2, r_3, \ldots r_n$ distances from a charge Q. The force on Q is given by the principle of superposition.

$$\vec{F} = \vec{F_1} + \vec{F_2} + \vec{F_3} + \cdots \vec{F_n}$$

$$= \frac{1}{4\pi\epsilon_0}\left(\frac{q_1 Q}{r_1^2}\hat{r}_1 + \frac{q_2 Q}{r_2^2}\hat{r}_2 + \frac{q_3 Q}{r_3^2}\hat{r}_3 + \cdots \frac{q_n Q \hat{r}_n}{r_n^2}\right)$$

$$\vec{F} = \frac{Q}{4\pi\epsilon_0}\left(\frac{q_1 \hat{r}_1}{r_1^2} + \frac{q_2 \hat{r}_2}{r_2^2} + \cdots \frac{q_n \hat{r}_n}{r_n^2}\right)$$

→ Let us define Electric field at a location of a test charge Q as

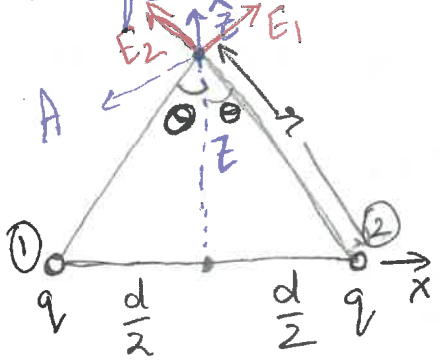
$$\boxed{\vec{E} = \frac{\vec{F}}{Q}}$$ It is vector with units $\frac{N}{C}$.

→ For several pt. charges the field is given as

$$\boxed{\vec{E} = \frac{1}{4\pi\epsilon_0}\sum_{i=1}^{n}\left[\frac{q_i}{r_i^2}\hat{r}_i\right]}$$

1(8) Find the electric field at a distance 'z' above the midpoint between 2 equal charges q, d apart.

Sol

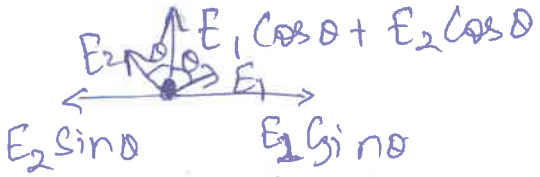


$\vec{E}_{net} = \vec{E}_1 + \vec{E}_2$

In z direction $E_1\cos\theta$ & $E_2\cos\theta$ components act.

In x direction $E_1\sin\theta$ & $E_2\sin\theta$ act in opposite directions

∵ q's are same $E_1 = E_2 = E$ At Pt. A, the fields are shown.



$\therefore E_{net}\big|_{z\ dirn.} = E_1\cos\theta + E_2\cos\theta = E\cos\theta + E\cos\theta$

$= 2E\cos\theta$

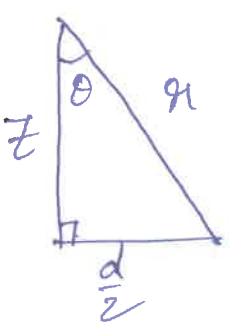
$E_{net}\big|_{x\ dirn} = E_1\sin\theta - E_2\sin\theta = E\sin\theta - E\sin\theta$

$= 0$

$\therefore E_{net\ total} = E_{net}\big|_{z\ dirn} = 2E\cos\theta\ \hat{z}$

$= 2\cdot\left(\frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\right)\cos\theta\ \hat{z}$

Now use



$r = \sqrt{z^2 + \frac{d^2}{4}}$

Also $\cos\theta = \frac{z}{r} = \frac{z}{\sqrt{z^2 + \frac{d^2}{4}}}$

$$\therefore \vec{E} = 2\left(\frac{1}{4\pi\epsilon_0}\frac{q}{(z^2+\frac{d^2}{4})}\cdot\frac{z}{\sqrt{z^2+\frac{d^2}{4}}}\right)\hat{z}$$

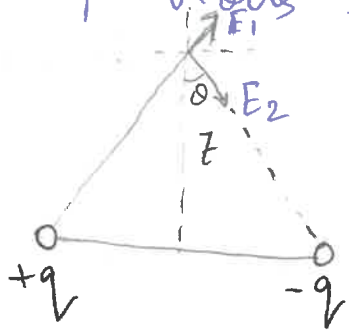
$$\boxed{\vec{E} = \frac{2qz}{4\pi\epsilon_0\left[z^2+\frac{d^2}{4}\right]^{3/2}}}$$

If we Consider $K = \frac{1}{4\pi\epsilon_0}$

$$\boxed{\vec{E} = \frac{2Kqz}{\left(z^2+\frac{d^2}{4}\right)^{3/2}}}$$

2) Now what happens if the charges are opposite in the previous scenario?

Sol



In this case, the "z" component Cancels each out and what remains is the x Component.

$$E_{net} = 2E\sin\theta \longrightarrow \frac{d}{2r} = \frac{d}{2\sqrt{z^2+\frac{d^2}{4}}}$$

$$\boxed{E = \frac{1}{4\pi\epsilon_0}\frac{qd}{\left(z^2+\frac{d^2}{4}\right)^{\frac{3}{2}}}\hat{x}}$$

→ Continuous Charge distribution:

Previously we considered discrete point charges. Now what happens if we have continuous charge distribution over a region? In such cases we use:

$$\vec{E}(r) = \frac{1}{4\pi\epsilon_0}\int \frac{dq}{r^2}\hat{r}$$

→ Charge densities:

$\lambda = \frac{\text{Charge}}{\text{length}}$ } line or linear charge density is given by charge per unit length.

$\sigma = \frac{\text{Charge}}{\text{area}}$ } Surface charge density is given by charge per unit area.

$\rho = \frac{\text{Charge}}{\text{volume}}$ } volume charge density is given by charge per unit volume.

Now, recall $\vec{dl}$ is line element, $\vec{da}$ is area element & $d\tau$ is volume element. Then

$$dq = \lambda dl$$
$$dq = \sigma da$$
$$dq = \rho d\tau$$

$$\because \quad \vec{E}(r) = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r^2} \hat{r}$$

We can write $\vec{E}$ depending on the distributed Charge as:

$$\vec{E}(r) = \frac{1}{4\pi\epsilon_0} \int \frac{\lambda\, dl}{r^2} \hat{r}$$ } for linear distributed Charges

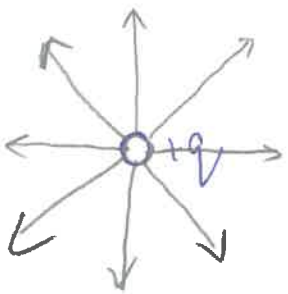
$$\vec{E}(r) = \frac{1}{4\pi\epsilon_0} \int \frac{\sigma\, da}{r^2} \hat{r}$$ } for charges distributed over a surface.

$$\vec{E}(r) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho\, d\tau}{r^2} \hat{r}$$ } for charges distributed over a volume

→ This is also Coulumb's law in terms of Electric field.

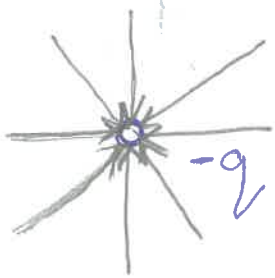
Field lines:

1)



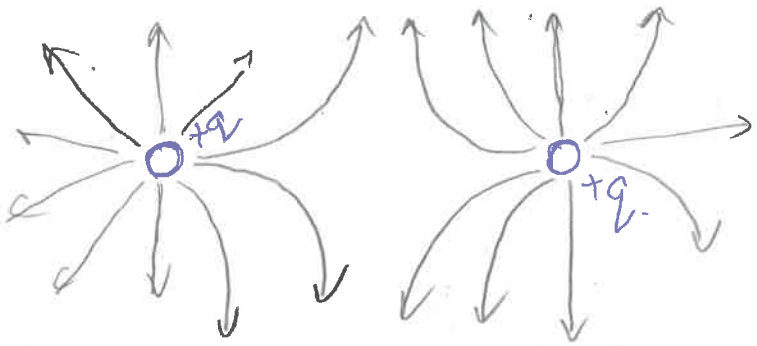
for positive point Charge +q

2)

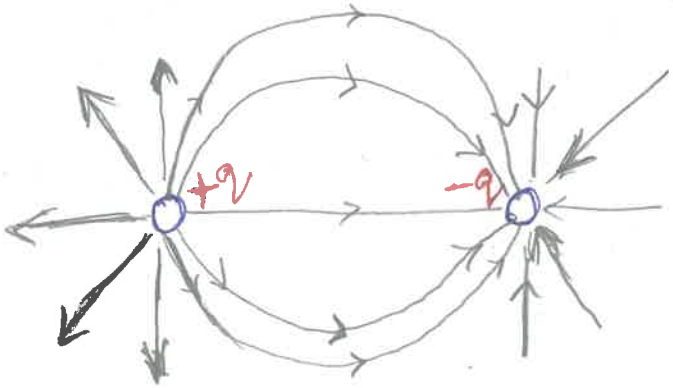


for negative point Charge -q.

3) Two equal similar charges:



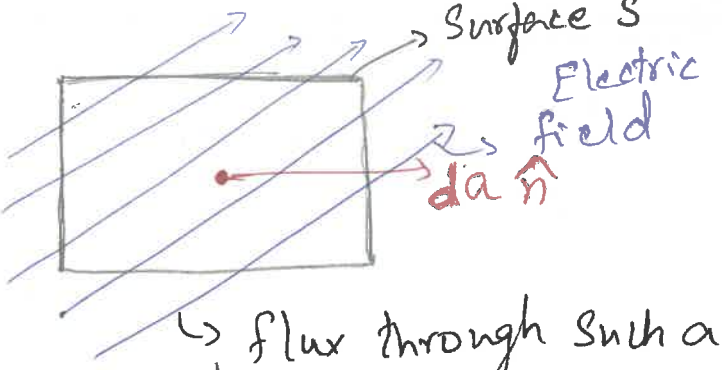
4) Two equal dissimilar charges:



Let us define flux as

$$\Phi_E = \int_S \vec{E} \cdot d\vec{a}$$

This is flux, which is a measure of number of field lines passing through a surface S



↳ flux through such a surface is given as

$$\Phi_E = \int_S \vec{E}\cdot d\vec{a}$$

→ or can also be written as $\oint \vec{E}\cdot d\vec{a}$

→ Now let us evaluate the flux for a point charge



→ Pt. Charge results in a spherical field in $\hat{r}$ direction.

For such a spherical distribution, area element $d\vec{a}$ in $\hat{r}$ is given by

$$d\vec{a} = r^2 \sin\theta\, d\theta\, d\phi\, \hat{r}$$

recall from area element in spherical coordinates

∴ Now the flux is

$$\oint \vec{E}\cdot d\vec{a} = \oint \left(\frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\hat{r}\right)\cdot\left(r^2\sin\theta\, d\theta\, d\phi\,\hat{r}\right)$$

$$= \oint \frac{1}{4\pi\epsilon_0}\frac{q}{r^2} r^2 \sin\theta\, d\theta\, d\phi$$

∵ θ & ϕ are independent we can break down the area integral into 2 seperate integrals.

$$= \int_{\theta=0}^{\pi}\int_{\phi=0}^{2\pi} \frac{q}{4\pi\epsilon_0} \sin\theta \, d\theta \, d\phi$$

$$= \int_{\theta=0}^{\pi} \frac{q}{4\pi\epsilon_0} \sin\theta \, d\theta \int_{\phi=0}^{2\pi} d\phi$$

$$= \frac{q}{4\pi\epsilon_0} \int_{\theta=0}^{\pi} \sin\theta \, d\theta \int_{0}^{2\pi} d\phi$$

$$= \frac{q}{4\pi\epsilon_0} \int_{\theta=0}^{\pi} \sin\theta \, d\theta \cdot (2\pi)$$

$$= \frac{q}{2\epsilon_0} \left[-\cos\theta \right]_0^{\pi}$$

$$= \frac{q}{2\epsilon_0} \left[-(-1-1) \right]$$

$$= \frac{2q}{2\epsilon_0}$$

$$= \frac{q}{\epsilon_0}$$

$$\therefore \boxed{\oint \vec{E} \cdot d\vec{a} = \frac{q}{\epsilon_0}}$$

$$\therefore \oint \vec{E} \cdot d\vec{a} = \frac{q}{\epsilon_0}$$ } This is for a pt. charge

If there are multiple charges enclosed by a surface then we have

$$\vec{E} = \sum_{i=1}^{n} \vec{E}_i$$ } Total field is sum of fields due to each charge

$$\therefore \oint \vec{E} \cdot d\vec{a} = \sum_{i=1}^{n} \oint \vec{E}_i \cdot d\vec{a} = \sum_{i=1}^{n} \frac{q_i}{\epsilon_0}$$

if $\sum_{i=1}^{n} q_i = Q_{enclosed}$

$$\oint \vec{E} \cdot d\vec{a} = \frac{Q_{enc}}{\epsilon_0}$$

→ This is called the Integral form of Gauss law.

The surface that encloses the charge Q_{enc} is called Gaussian surface.

Divergence Theorem

It states that "Integral of a derivative over a region is equal to the value of the function at the boundary"

→ This is also known as Gauss divergence theorem, or Greens theorem.

$$\int_V (\nabla \cdot \vec{T})\, d\tau = \oint \vec{T} \cdot d\vec{a}$$

↳ Divergence Theorem.

Using divergence theorem on Gauss law:

$$\Phi = \oint \vec{E} \cdot d\vec{a} = \int_V (\nabla \cdot \vec{E})\, d\tau$$

Use $dq = \rho\, d\tau$ (where $\rho = \frac{Q}{V}$ charge density)

$$\therefore Q_{enc} = \int \rho\, d\tau$$

$$\therefore \oint \vec{E} \cdot d\vec{a} = \frac{Q_{enc}}{\epsilon_0} = \frac{\int_V \rho\, d\tau}{\epsilon_0} = \int_V \left(\frac{\rho}{\epsilon_0}\right) d\tau$$

& $\oint \vec{E} \cdot d\vec{a} = \int_V (\nabla \cdot \vec{E})\, d\tau$ } from divergence

These 2 give the flux ∴ they can be equated

$$\therefore \int_V \left(\frac{\rho}{\epsilon_0}\right) d\tau = \int_V (\nabla \cdot \vec{E})\, d\tau$$

$$\Rightarrow \boxed{\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}}$$

→ This is the differential form of Gauss's law.

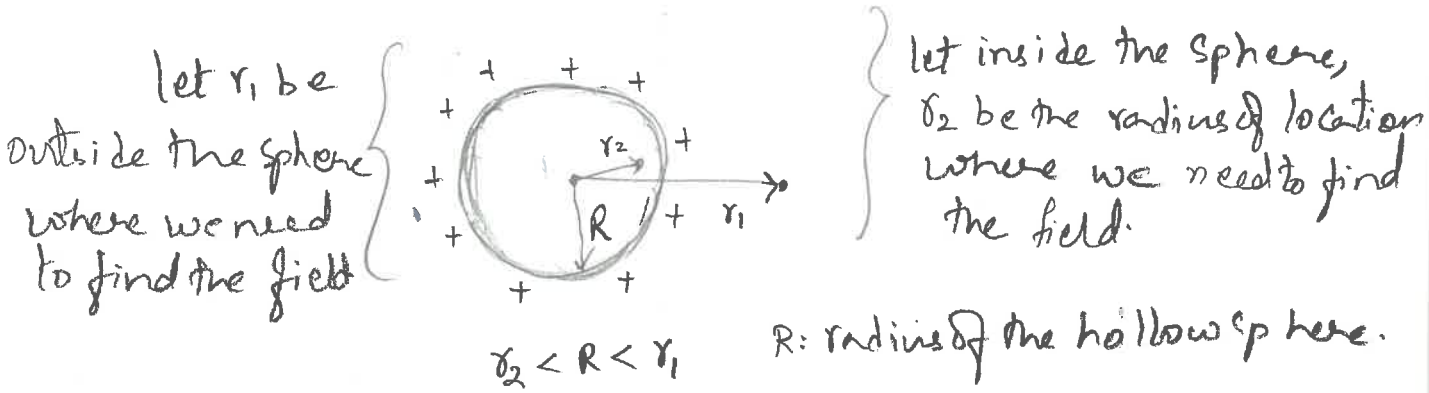
Applications of Gauss's law

$$\oint \vec{E} \cdot d\vec{a} = \frac{Q_{enc}}{\epsilon_0} \quad \text{or} \quad \vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

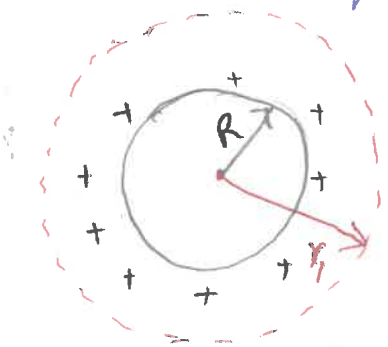
Different versions of Gauss law.

→ Find the field outside & inside of a Uniformly Charged Hollow Sphere.

Sol First draw the given object



a) Outside the hollow sphere



Draw the gaussian surface (spherical) at r_1). Now for this surface apply Gauss's law.

$$\oint \vec{E} \cdot d\vec{a} = \frac{Q_{enc}}{\epsilon_0}$$

$$\oint \vec{E} \cdot r_1^2 \sin\theta \, d\theta \, d\phi = \frac{Q_{enc}}{\epsilon_0}$$

$$\vec{E} r_1^2 \int_0^{\pi} \sin\theta \, d\theta \int_0^{2\pi} d\phi = \frac{Q_{enc}}{\epsilon_0}$$

$$\vec{E} \cdot r_1^2 \cdot 2 \cdot 2\pi = \frac{Q_{enc}}{\epsilon_0}$$

let $q = Q_{enc}$ be the total charge on the hollow sphere

$$\therefore \vec{E} = \frac{q}{4\pi\epsilon_0 r_1^2} \hat{r}$$

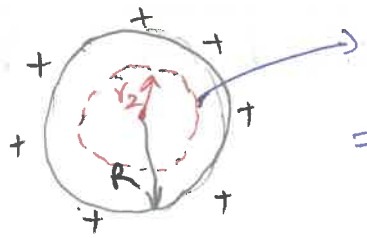
∴ at r_1 which is outside the hollow sphere the field behaves as if there is a pt. charge at the Center of hollow sphere.

$$\vec{E}\Big|_{\text{outside the surface}} = \frac{q}{4\pi\epsilon_0 R^2} \hat{r}$$

b) Inside the hollow sphere

∵ at $r_2 < R$ there is no charge

$\vec{E} = 0$ inside the hollow sphere.

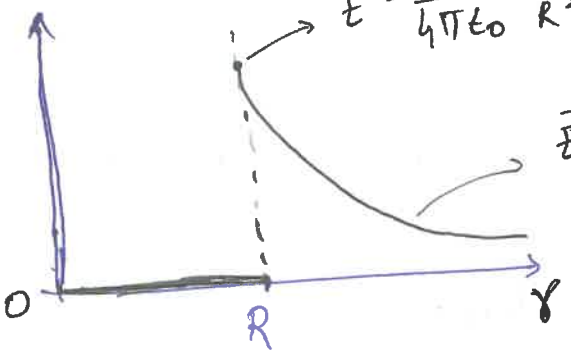


$\rightarrow q = 0$

$\Rightarrow E = 0$ inside

$\vec{E} = \frac{1}{4\pi\epsilon_0}\frac{q}{R^2}$

$\therefore \vec{E}(r)$



$\vec{E} \propto \frac{1}{r^2}$ for $r > R$

→ Find the field Inside & outside of uniformly charged Solid sphere

Sol Similar to previous problem, solve $\vec{E}$ for outside the Sphere first.



$r_1 > R$

$\oint E \cdot da = \frac{Q_{enc}}{\epsilon_0}$ let $Q_{enc} = Q$

$E \cdot 4\pi r_1^2 = \frac{Q}{\epsilon_0}$ } once again the area integral gives $4\pi r_1^2$

$$E = \frac{Q}{4\pi\epsilon_0 r_1^2}$$

As expected, the field outside the sphere is behaving as if a point charge is concentrated at the center.

→ Now for the field inside the sphere:



∵ It is a uniformly distributed charge, we use

$$\rho = \frac{q_{enc}}{Vol}$$

at $r=R$ $\quad \rho = \dfrac{Q}{\frac{4}{3}\pi R^3}$

at $r=r_2$ $\quad \rho = \dfrac{q_{enc}}{\frac{4}{3}\pi r_2^3}$

$$\therefore \boxed{q_{enc} = \frac{Q r_2^3}{R^3}}$$

∴ Using Gauss's law

$$\oint \vec{E} \cdot d\vec{a} = \frac{q_{enc}}{\epsilon_0} = \frac{Q r_2^3}{R^3 \epsilon_0}$$

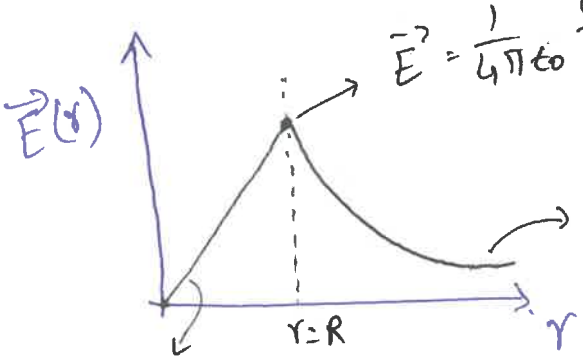
$$\vec{E} \cdot 4\pi r_2^2 = \frac{Q r_2^3}{R^3 \epsilon_0}$$

$$\boxed{\vec{E} = \frac{Q r_2}{4\pi\epsilon_0 R^3} \hat{r}}$$

On the surface of the solid sphere

$$\vec{E} = \frac{1}{4\pi\epsilon_0}\frac{Q}{R^2}\hat{r}$$

$\therefore$



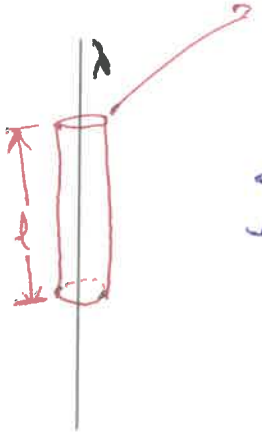
$$\vec{E} = \frac{1}{4\pi\epsilon_0}\frac{Q}{R^2}$$

$$\vec{E} = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2} \text{ for } r > R.$$

$$\vec{E} = \frac{1}{4\pi\epsilon_0}\frac{Q}{R^3}\vec{r} \text{ (for } r < R)$$

→ for Uniformly distributed charged infinitely long wire:

Finding the field for such a long wire using Gauss's law:



$$\oint \vec{E}\cdot d\vec{a} = \frac{Q_{enc}}{\epsilon_0}$$

$$\vec{E}\cdot\int r\,d\phi\,dz = \frac{\lambda\cdot l}{\epsilon_0}$$

$$\vec{E}\cdot r\int_0^{2\pi} d\phi\int_0^{l} dz = \frac{\lambda l}{\epsilon_0}$$

$$\vec{E}\cdot r\cdot 2\pi\cdot l = \frac{\lambda l}{\epsilon_0} \Rightarrow \vec{E} = \frac{\lambda}{2\pi\epsilon_0 r}\hat{r} \text{ or } \frac{2k\lambda}{r}\hat{r}$$

where $k = \frac{1}{4\pi\epsilon_0}$

→ Field for ∞ long sheet:

∞ long sheet with σ charge density.



→ shape of the Gaussian Surface

Area is "A"

$$\oint \vec{E} \cdot d\vec{a} = \frac{Q_{enc}}{\epsilon_0}$$

$$2A\vec{E} = \frac{\sigma \cdot A}{\epsilon_0}$$

$$\boxed{\vec{E} = \frac{\sigma}{2\epsilon_0}\hat{n}}$$